

CIS241

System-Level Programming and Utilities

File Transfers

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Based on material provided by Erin Carrier, Austin Ferguson, and Katherine Bowers

Lots of options!

Ones we'll talk about:

- scp
- sftp
- rsync

SCP



SCP

Remember `ssh` - Secure SHell

`scp` - Secure CoPy

Recall - copying files locally: `cp source destination`

`scp` is similar, but works remotely!

Information required:

- Host we are copying to/from
- Username we'll use on that host

SCP

```
scp [[user@]src_host:]src_path [[user@]dest_host:]dest_path
```

- Note that items in [] can sometimes be omitted
- Example, transfer local file to EOS

- ```
scp local_file username@eos01.cis.gvsu.edu:~/my_dir
```

# SCP directedness

**scp doesn't care which systems we are moving files to/from**

**The last example copied a local file to the server**

- `scp local_file username@eos01.cis.gvsu.edu:~/my_dir`

**What if we want to copy a file from server to laptop?**

- `scp username@eos01.cis.gvsu.edu:~/my_dir/file .`

**Why stop there? We could also copy files directly from one server to another, all from our local machine!**

# SFTP

## Secure File Transfer Protocol

`sftp username@address`

### Common commands:

- `get file`
- `put file`
- Limited subset of terminal commands (e.g., `ls`)
- Can interact with local (your computer) terminal using `!` in front
  - e.g., `!ls` lists the files in your current directory

# SFTP

## SFTP Commands

Cheat Sheet Series

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| Command        | Description                                                      |
|----------------|------------------------------------------------------------------|
| <b>?</b>       | List of all commands or explain a command, eg ? get              |
| <b>!</b>       | Leave the environment temporarily *                              |
| <b>cd</b>      | Change the active directory on the remote host                   |
| <b>chmod</b>   | Change the permissions of files on the remote host               |
| <b>chown</b>   | Change the owner of files on the remote host                     |
| <b>dir</b>     | List the contents of the current directory on the remote host    |
| <b>exit</b>    | Close the connection and leave SFTP                              |
| <b>get</b>     | Copy a file from the remote host to the local computer           |
| <b>help</b>    | Same as ?                                                        |
| <b>lcd</b>     | Change the active directory on the local computer                |
| <b>lls</b>     | List the contents of the current directory on the local computer |
| <b>lmkdir</b>  | Create a directory on the local computer                         |
| <b>ln</b>      | Create a symbolic link for a file on the remote host             |
| <b>lpwd</b>    | Show the present working directory on the local computer         |
| <b>ls</b>      | Same as dir                                                      |
| <b>lumask</b>  | Change the local umask value                                     |
| <b>mkdir</b>   | Create a directory on the remote host                            |
| <b>put</b>     | Copy a file from the local computer to the remote host           |
| <b>pwd</b>     | Show the present working directory on the remote host            |
| <b>quit</b>    | Same as exit                                                     |
| <b>rename</b>  | Rename a file on the remote host                                 |
| <b>rm</b>      | Delete a file on the remote host                                 |
| <b>rmdir</b>   | Remove an empty directory on the remote host                     |
| <b>symlink</b> | Same as ln                                                       |
| <b>version</b> | Show the SFTP version                                            |

# Which one?

- **SCP is older, SFTP is newer**
- **Both get the job done *fine***
- **However**
  - SFTP gives you a bit *more*
    - Such as resumable transfers, file system access, etc.
- **Both still better than FTP!**
  - Why?



# RSYNC



**rsync is like a fancier scp**

**It checks to make sure there are differences between the files**

- If there are differences, it only copies them, not the whole file

**Command format is very similar to scp:**

- `rsync [[user@]src_host:]src_path [[user@]dest_host:]dest_path`

**Note that rsync has a ton of options!**

- Automatic compression
- Progress bars
- Partial files (in case of restarts)
- Much more!

# rsync

<https://www.digitalocean.com/community/tutorials/how-to-use-rsync-to-sync-local-and-remote-directories>